CLH-MAX: A Lattice-Based KEM Proposal over the Prime Cyclotomic Ring $\mathbb{Z}_q[X]/\Phi_{257}(X)$

Santiago López Heinzen
Independent researcher
lopezantiago79@gmail.com

October 2026 (version 2.0.1)

Abstract

We propose CLH-MAX, a Module-LWE-based key encapsulation mechanism (KEM) in the style of Kyber/ML-KEM in which the ring $\mathbb{Z}_q[X]/(X^{256}+1)$ is replaced by $R_q = \mathbb{Z}_q[X]/\Phi_{257}(X)$, the ring of the prime cyclotomic field $\mathbb{Q}(\zeta_{257})$. For primes q that are primitive roots modulo 257 the polynomial Φ_{257} is irreducible modulo q, so $R_q \cong \mathbb{F}_{q^{256}}$ is a field. We specify the CPA-secure public-key encryption layer, its Fujisaki–Okamoto transform to an IND-CCA2 KEM (random-oracle model), and three parameter sets of module rank $2, 3, 4$. We give a game-hopping proof that IND-CPA security reduces to the decision Module-LWE problem over R_q, an exact analysis of the decryption failure probability (which corrects the naïve coefficient-variance count for this ring and shows that the ML-KEM modulus $q = 3329$ is not adequate for rank 4), a choice of the modulus per module rank so that $\delta < 2^{-128}$ for all three sets, lattice-estimator outputs for key- and message-recovery instances (identical to ML-KEM-768 for $k = 3$, within 0.2 bits of the weakest ML-KEM-512 instance for $k = 2$, and about 2 bits below ML-KEM-1024 for $k = 4$), and a discussion of arithmetic cost, where the absence of a negacyclic NTT is the main disadvantage. We make no claim of higher concrete security than ML-KEM; the contribution is a fully specified alternative ring instantiation together with its correctness and reduction analysis, reproducible scripts, and a clear list of open problems.

Status. This is an unrefereed technical note; it has not been peer-reviewed. Its claims are limited to what is proved or computed in the text and in the accompanying scripts.

1 Introduction

ML-KEM (FIPS 203 [10]) is based on Module-LWE over $\mathbb{Z}_{3329}[X]/(X^{256}+1)$, a power-of-two cyclotomic ring chosen because it admits a fast (incomplete) number-theoretic transform (NTT). The choice of ring in structured-lattice cryptography is a design parameter that is not fully explored: other cyclotomic rings have been proposed, e.g. the prime-conductor rings $\mathbb{Z}_q[X]/\Phi_{n+1}$ ($n+1$ prime) used in the NIST round-2 candidates Round2 and Round5 [8]. This note studies one concrete alternative, conductor 257 (the Fermat prime $F_3 = 2^8+1$), for which the ring degree is $\varphi(257) = 256$, equal to that of ML-KEM, so module ranks, key sizes and noise parameters can be compared directly.

Contributions. (1) A complete specification of CLH-MAX: PKE, FO-based KEM, three parameter sets (Sections 3–4). (2) A proof that IND-CPA security reduces to two decision-MLWE instances over R_q, with the explicit simulators (Section 5, Appendix B). (3) An exact computation of the decryption failure probability including the ring-specific coefficient count for Φ_{257}; it shows that with

the ML-KEM modulus $q = 3329$ the rank-4 set does *not* reach 2^{-128}, which we repair by choosing the modulus per rank (Sections 6,7, Appendices A,C). (4) A cost analysis of multiplication in R_q (Section 8).

Non-claims. We do not claim that CLH-MAX is more secure than ML-KEM, that known attacks on ML-KEM fail to apply to a field ring, or that worst-case reductions give concrete security at these parameters. Section 5 explains why.

Related work. Ring-LWE over general cyclotomic rings was introduced in [9], including the worst-case reduction for arbitrary cyclotomic fields. Kyber [3] and ML-KEM use module structure over $\mathbb{Z}_q[X]/(X^{256}+1)$; the module variant over $\mathbb{Z}_q[X]/\Phi_{257}$ studied here follows the same template. Prime-conductor cyclotomic rings were already used in Round5 [8], which is based on ring learning with rounding; we make no claim of novelty for the ring itself. Subfield and structural attacks are discussed in [1, 4, 5, 6].

2 Preliminaries

Notation. $N = 256$, q prime, $R_q = \mathbb{Z}_q[X]/\Phi_{257}(X)$ with $\Phi_{257} = X^{256} + \cdots + X + 1$. Elements are identified with coefficient vectors in $\mathbb{Z}_q^{256}$. For $k \geq 1$, vectors and matrices over R_q are written with the module rank k as dimension. ψ_η is the centered binomial distribution on $\mathbb{Z}$: $\sum_{i=1}^{\eta}(a_i - b_i)$ with $a_i, b_i \leftarrow \{0,1\}$ uniform; it has variance $\eta/2$. For $\eta = 2$ the variance is 1 (we use $\eta = 2$ throughout, so $\sigma^2 = 1$). χ denotes ψ_2 applied coefficient-wise.

Definition 1 (Decision MLWE)**.** *Let $n, m \geq 1$.* $\mathrm{MLWE}_{n,m,\chi}$ *asks to distinguish $(B, Bs+e)$ from (B, w), where $B \leftarrow U(R_q^{m\times n})$, $s \leftarrow \chi^n$, $e \leftarrow \chi^m$, $w \leftarrow U(R_q^m)$.* $\mathrm{Adv}^{\mathrm{MLWE}}_{n,m,\chi}(\mathcal{D})$ *is* $|\Pr[\mathcal{D}(B, Bs+e) = 1] - \Pr[\mathcal{D}(B, w) = 1]|$.

Definition 2 (PKE, IND-CPA)**.** *A PKE is* $(\mathsf{KeyGen}, \mathsf{Enc}, \mathsf{Dec})$*. In the IND-CPA game the challenger samples (pk, sk) and a bit b; the adversary picks m_0, m_1 of equal length, receives* $\mathsf{Enc}(pk, m_b)$ *and outputs b'. The advantage is* $|\Pr[b' = b] - \frac{1}{2}|$.

Definition 3 (KEM, IND-CCA2)**.** *A KEM is* $(\mathsf{KeyGen}, \mathsf{Encaps}, \mathsf{Decaps})$*, with* $\mathsf{Encaps}(pk) \to (c, K)$*. In the IND-CCA2 game the adversary receives (c^*, K_b) with K_0 the real key and K_1 uniform, and may query* Decaps *on any $c \neq c^*$; advantage* $|\Pr[b' = b] - \frac{1}{2}|$.

3 The ring

Proposition 4. *Let $q \neq 257$ be prime. Then Φ_{257} is irreducible over $\mathbb{F}_q$ if and only if* $\mathrm{ord}_{257}(q) = 256$*, i.e. q is a primitive root modulo 257. In that case $R_q \cong \mathbb{F}_{q^{256}}$ and $q\mathcal{O}_K$ is a prime ideal of $\mathcal{O}_K = \mathbb{Z}[\zeta_{257}]$.*

Proof. For $\gcd(q, m) = 1$ all irreducible factors of Φ_m over $\mathbb{F}_q$ have degree $\mathrm{ord}_m(q)$; here $\deg \Phi_{257} = \varphi(257) = 256$. The statement about $q\mathcal{O}_K$ is the standard description of inert primes in cyclotomic fields. □

Direct computation gives $\mathrm{ord}_{257}(q) = 256$ for $q = 2531$ ($\equiv 218$), 3329 ($\equiv 245$), 3557 ($\equiv 216$) and 12289 ($\equiv 210 \pmod{257}$). Among the primes below 20000 there are 1142 moduli with this property. Not every NTT-friendly prime works: $\mathrm{ord}_{257}(7681) = 128$, so Φ_{257} splits into two factors of degree 128 modulo 7681. The moduli used in this paper are $q = 2531$, 3329 and 3557, chosen per module rank in Section 7; 3329 is the ML-KEM modulus.

Remark 5 (Comparison with ML-KEM). For $q = 3329$ one has $q \equiv 1 \pmod{256}$ but $q \not\equiv 1 \pmod{512}$ ($3329 \bmod 512 = 257$, $\mathrm{ord}_{512}(3329) = 2$). Hence $X^{256}+1$ factors modulo 3329 into 128 *quadratic* factors, not 256 linear ones; this is why ML-KEM uses an incomplete, 7-layer NTT. $q = 12289$, which splits $X^{256}+1$ completely, is not an ML-KEM modulus.

Remark 6 (Subfields). $\mathbb{F}_{q^{256}}$ is not "subfield-free": it contains the subfields $\mathbb{F}_{q^{2^j}}$, $0 \le j \le 7$, and $\mathrm{Gal}(\mathbb{F}_{q^{256}}/\mathbb{F}_q)$ is cyclic of order 256; likewise $\mathbb{Q}(\zeta_{257})$ has a cyclic Galois group of order 256 and a subfield of every degree 2^j. What differs from $X^{256}+1$ is only that R_q does not decompose as a product of smaller rings. Whether this has any security consequence is open.

Multiplication. Let $a, b \in R_q$ with coefficient vectors of length 256 (degree ≤ 255). Then $ab = \sum_{i=0}^{510} c_i X^i$ over $\mathbb{Z}_q$. Since $\Phi_{257} \mid X^{257}-1$, first fold $p_i = \sum_{j+l \equiv i \ (257)} a_j b_l$ for $0 \le i \le 256$; then $ab \bmod \Phi_{257}$ has coefficients $p_i - p_{256}$ for $0 \le i \le 255$ (because $X^{256} \equiv -(X^{255} + \cdots + 1)$). Hence a ring multiplication is a length-257 cyclic convolution followed by 256 subtractions.

4 Scheme

Fix $\eta = 2$, module rank $k \in \{2,3,4\}$ and the modulus q_k of Table 1: $q_2 = 2531$, $q_3 = 3329$, $q_4 = 3557$ (Section 7); we write q for q_k. Let $\mathrm{Enc}_1(m) = \lceil q/2 \rfloor \cdot m$ embed $m \in \{0,1\}^{256}$ coefficient-wise into R_q and $\mathrm{Dec}_1(x)_i = 1$ iff the centered representative of x_i lies in $[\lceil q/4 \rceil, \lfloor 3q/4 \rfloor]$.

PKE.

- KeyGen: $A \leftarrow U(R_q^{k\times k})$; $s, e \leftarrow \chi^k$; $t = As + e$. $pk = (A, t)$, $sk = s$.
- $\mathsf{Enc}(pk, m; r)$: coins determine $r, e_1 \leftarrow \chi^k$, $e_2 \leftarrow \chi$; $u = A^T r + e_1$; $v = t^T r + e_2 + \mathrm{Enc}_1(m)$. Output (u, v).
- $\mathsf{Dec}(sk, (u, v))$: $m' = \mathrm{Dec}_1(v - s^T u)$.

Correctness. $v - s^T u = \mathrm{Enc}_1(m) + E$ with $E = e^T r + e_2 - s^T e_1$; decryption succeeds iff every coefficient of E has centered absolute value $< \lfloor q/4 \rfloor$ (up to rounding of $q/2$).

KEM (FO with implicit rejection). With hash functions H, G modelled as random oracles and following [7]:

- KeyGen: $(pk, s) \leftarrow \mathsf{PKE.KeyGen}$, $z \leftarrow \{0,1\}^{256}$; $sk_{\mathrm{KEM}} = (s, pk, H(pk), z)$.
- $\mathsf{Encaps}(pk)$: $m \leftarrow \{0,1\}^{256}$; $(\bar{K}, r) = G(m \,\|\, H(pk))$; $c = \mathsf{Enc}(pk, m; r)$; output $(c,\ K = H(\bar{K} \,\|\, H(c)))$.
- $\mathsf{Decaps}(sk, c)$: $m' = \mathsf{Dec}(s, c)$; $(\bar{K}', r') = G(m' \| H(pk))$; if $c = \mathsf{Enc}(pk, m'; r')$ (constant-time comparison) return $H(\bar{K}' \| H(c))$, else return $H(z \| H(c))$ (implicit rejection, also in constant time).

By [7], IND-CCA2 security of this KEM in the random-oracle model follows from IND-CPA security of the PKE together with a sufficiently small decryption failure probability δ; the explicit bound of [7] should be consulted for the concrete loss in terms of the number of oracle queries. We make no quantum-random-oracle claim.

Sizes (no compression). All three moduli are below 4096, so 12-bit packing applies: $|pk| = 384k + 32$ B, $|c| = 384(k+1)$ B, $|sk_{\mathrm{KEM}}| = 384k + |pk| + 64$ B. Ciphertext compression as in ML-KEM is not specified here and would change both δ and ciphertext size.

Set	k	$n = 256k$	q	$\lvert pk\rvert$	$\lvert c\rvert$	$\lvert sk_{\mathrm{KEM}}\rvert$
CLH-MAX-512	2	512	2531	800	1152	1632
CLH-MAX-768	3	768	3329	1184	1536	2400
CLH-MAX-1024	4	1024	3557	1568	1920	3168

Table 1: Parameter sets and sizes in bytes; $\eta = 2$, no ciphertext compression.

5 Security

Theorem 7 (IND-CPA)**.** *For every adversary $\mathcal{A}$ against the PKE there are distinguishers $\mathcal{B}_1, \mathcal{B}_2$ with essentially the running time of $\mathcal{A}$ such that*

$$\mathrm{Adv}^{\text{IND-CPA}}(\mathcal{A}) \le \mathrm{Adv}^{\mathrm{MLWE}}_{k,k,\chi}(\mathcal{B}_1) + \mathrm{Adv}^{\mathrm{MLWE}}_{k+1,k,\chi}(\mathcal{B}_2).$$

Proof. *Game* G_0 is the real IND-CPA game. *Game* G_1 is G_0 with t replaced by a uniform $t^* \in R_q^k$. $\mathcal{B}_1$, given an MLWE challenge (A, t^*) with k samples and secret s, runs $\mathcal{A}$ on $pk = (A, t^*)$, plays the challenger honestly for the rest and outputs 1 iff $\mathcal{A}$ wins. If $t^* = As + e$ this is exactly G_0, otherwise G_1, so $|\Pr[G_0] - \Pr[G_1]| = \mathrm{Adv}^{\mathrm{MLWE}}_{k,k,\chi}(\mathcal{B}_1)$. *Game* G_2 replaces (u, v) by uniform elements of $R_q^k \times R_q$. In G_1 the matrix $B = \binom{A^T}{t^T} \in R_q^{(k+1)\times k}$ is uniform and

$$\begin{pmatrix} u \\ v - \mathrm{Enc}_1(m_b) \end{pmatrix} = B\, r + \begin{pmatrix} e_1 \\ e_2 \end{pmatrix}$$

is an MLWE sample with secret r and $k+1$ rows. $\mathcal{B}_2$, given (B, w), parses B into A^T and t^T, runs $\mathcal{A}$ on $pk = (A, t)$, and after the challenge messages are chosen returns $\mathcal{A}$ the ciphertext $(w_{1..k},\ w_{k+1} + \mathrm{Enc}_1(m_b))$. Real samples give G_1, uniform ones G_2. In G_2 the ciphertext is independent of b so $\Pr[G_2] = \frac{1}{2}$. Combining, $\Pr[G_0] - \frac{1}{2} \le \varepsilon_1 + \varepsilon_2$. □

Remark 8 (Why no "axiom"). The theorem is a relative statement. It does *not* give $\mathrm{Adv}^{\text{IND-CPA}} = 0$: an unbounded adversary recovers s from (A, t) (the support of (s, e) is far smaller than q^{256k}), so no sound assumption can set the MLWE advantage of *all* distinguishers to zero. Security is computational and rests on the hardness of MLWE for efficient adversaries. Any mechanised proof should formalise the inequality of Theorem 7, not an unconditional-hardness axiom.

Worst-case reductions. Lyubashevsky–Peikert–Regev [9] give a quantum reduction from approximate worst-case SVP on ideal lattices to RLWE for *any* cyclotomic field, hence including $\mathbb{Q}(\zeta_{257})$, but for discrete Gaussian errors in the canonical embedding with width $\alpha q \ge \omega(\sqrt{\log n})$ and with moduli and approximation factors that are polynomial in n with non-trivial exponents. The parameters used here ($q \le 3557$, centered binomial $\eta = 2$, $n = 256$) lie outside that regime, exactly as for ML-KEM, so the reductions provide asymptotic motivation only and no concrete bits of security. Moreover, for prime-power cyclotomic fields (which includes conductor 257) there are quantum algorithms for ideal-SVP with approximation factors $\exp(\tilde{O}(\sqrt{n}))$ in certain settings [5, 6]; these do not break RLWE at the parameters considered, but they caution against reading worst-case statements as guarantees.

Concrete security. Attacks on (M)LWE (primal uSVP, BDD, dual, hybrid) as implemented in the lattice-estimator [2] see only the instance parameters $(n, m, q, \chi_s, \chi_e)$; they do not depend on the ring structure. We ran `LWE.estimate` (default settings, October 2026) on two instances per

parameter set: *key recovery* ($n = 256k$, $m = n$, secret and error ψ_2) and *message recovery* (secret r, errors (e_1, e_2): $n = 256k$, $m = 256(k + 1)$), and, with the same call, on the corresponding ML-KEM instances (ML-KEM-512: ψ_3 secret and error for key recovery, ψ_3 secret and ψ_2 error for message recovery). Table 2 reports the output unchanged. For ML-KEM-512 the message-recovery

Scheme	Instance	BKW	uSVP	BDD	Dual	DH	β_{DH}
ML-KEM-512	key (m=512)	178.8	143.8	140.2	149.9	139.7	387
ML-KEM-512	message (m=768)	178.8	149.6	144.9	146.2	136.6	376
CLH-512 (2531)	key	173.0	141.5	137.9	147.6	136.8	374
CLH-512 (2531)	message	172.9	141.5	137.9	147.6	136.8	374
ML-KEM-768 = CLH-768 (3329)	key	238.3	204.9	201.0	214.3	196.4	589
ML-KEM-768 = CLH-768 (3329)	message	238.1	204.9	201.0	214.3	196.4	589
ML-KEM-1024	key	315.0	275.1	270.7	288.5	262.3	823
ML-KEM-1024	message	310.5	275.1	270.7	288.5	262.3	823
CLH-1024 (3557)	key	311.2	273.2	268.8	286.4	260.4	817
CLH-1024 (3557)	message	311.1	273.2	268.8	286.4	260.4	817

Table 2: lattice-estimator output, $\log_2$(operations); one call and one cost model for all rows. DH = dual hybrid, the cheapest attack in every row. For $k = 3$ the instances are literally identical for CLH-MAX and ML-KEM.

instance is the weakest ($2^{136.6}$, dual hybrid), cheaper than key recovery ($2^{139.7}$). Taking for each scheme the cheapest attack over both instances: for $k = 2$, CLH-MAX-512 gives $2^{136.8}$ against $2^{136.6}$ for ML-KEM-512 (+0.2 bits); for $k = 3$ both give $2^{196.4}$; for $k = 4$, CLH-MAX-1024 gives $2^{260.4}$ against $2^{262.3}$ for ML-KEM-1024 (-1.9 bits). For CLH-MAX the two instances coincide up to rounding, because secret and error share the same distribution. We draw no ranking conclusion from differences of this size: in this model CLH-MAX matches ML-KEM to within about two bits, and the ring itself plays no role in the estimates. Limitations: the estimates are for generic MLWE, they use the independence of the coordinates of (s, e) built into the estimator, and we assign no NIST category. Known structural attacks (subfield attacks on overstretched NTRU [1, 4]) target large-modulus NTRU-type instances and are not known to apply to MLWE at these moduli in either ring; Remark 6 explains why the field structure of R_q is not obviously free of exploitable structure.

6 Correctness: decryption failure probability

Lemma 9. *For $a, b \in R_q$ with independent coefficients, every coefficient of $ab \in R_q$ is a signed sum of* 510 *products $a_j b_l$ of distinct index pairs, except coefficient* 255 *which is a signed sum of* 511.

The proof is a direct count of index pairs (Appendix A). This is roughly twice the 256 terms of a negacyclic product, so the variance of a product coefficient is about $2\sigma^4 \cdot 256$ rather than $256\sigma^4$.

For module rank k, the coefficient E_i of $e^T r - s^T e_1 + e_2$ is therefore a sum of $2k \cdot 511$ products of two independent ψ_2 samples (a conservative count) plus one ψ_2 sample, with variance $\sigma_E^2 = 1022k + 1$. Following the usual heuristic of treating the summands as independent, we compute the *exact* distribution by convolution and bound $\delta \le 256 \cdot \Pr[|E_i| \ge \lfloor q/4 \rfloor]$. The script is in Appendix C. All three sets satisfy $\delta < 2^{-128}$ with a margin of at least 5 bits under this heuristic. A rigorous bound would have to account for the dependence between products that share a factor; we do not provide one.

Set	k	q	$\lfloor q/4 \rfloor$	σ_E^2	$\log_2 \Pr[\lvert E_i \rvert \geq \lfloor q/4 \rfloor]$	$\log_2 \delta$
CLH-MAX-512	2	2531	632	2045	−141.5	−133.5
CLH-MAX-768	3	3329	832	3067	−163.9	−155.9
CLH-MAX-1024	4	3557	889	4089	−141.7	−133.7

Table 3: Decryption failure probability, no ciphertext compression, $\eta = 2$.

7 Parameter selection

Two effects pull in opposite directions. Decreasing q with fixed ψ_2 noise increases the cost of lattice attacks, but it also shrinks the decoding threshold $\lfloor q/4 \rfloor$ and hence increases δ. Because of Lemma 9 the variance of a product coefficient is about twice that of a negacyclic ring, so with the ML-KEM modulus $q = 3329$ the rank-4 set has $\delta \approx 2^{-117}$, and raising the key noise of the rank-2 set to ML-KEM-512's $\eta_1 = 3$ (with $r \leftarrow \psi_3$, $e_1, e_2 \leftarrow \psi_2$) gives $\delta \approx 2^{-123}$; neither meets 2^{-128}. We therefore scanned all primes $2000 < q < 20000$ with $\mathrm{ord}_{257}(q) = 256$, for $k \in \{2, 4\}$ and $\eta \in \{2, 3\}$, using (a) a primal-attack estimate of the uSVP block size, which reproduces exactly the four uSVP block sizes of our reference estimator runs, and (b) a Chernoff bound on δ (conservative, about 5 bits above the exact value). The finalists were then confirmed with the lattice-estimator and with the exact computation of Section 6: For $k = 3$, $q = 3329$ already matches ML-KEM-768 and gives

k	q	β_{uSVP}	DH (key)	$\log_2 \delta$	status
2	3329	382	132.8	−231.0	superseded
2	**2531**	398	136.8	−133.5	**chosen**
2	2663	395	136.2	−148.0	alternative
4	3329	874	262.3	−116.7	fails δ
4	**3557**	867	260.4	−133.7	**chosen**
4	3617	865	259.5	−138.4	alternative

Table 4: Candidate moduli ($\eta = 2$). For the alternatives only the key-recovery instance was run with the estimator. DH = dual hybrid, $\log_2$ operations.

$\delta \approx 2^{-156}$, so it is kept. For $k = 2, 4$ we choose the smaller modulus, which keeps at least 5 bits of margin in δ and is closest to ML-KEM in the estimator; the larger alternatives give up about one bit of security for 5–15 additional bits of margin. The choice is a judgement, not an optimum, and a different weighting of the two criteria is equally defensible.

8 Arithmetic

By Section 3, multiplication in R_q is a length-257 cyclic convolution followed by a correction. Schoolbook multiplication costs $256^2 = 65{,}536$ coefficient products per ring multiplication; KeyGen needs k^2 ring products for As and Enc needs $k^2 + k + k$ more. A fast NTT route requires a transform of length ≥ 511 covering the linear convolution, i.e. $512 \mid q-1$. None of the moduli used here satisfies this: $2530 = 2 \cdot 5 \cdot 11 \cdot 23$, $3328 = 2^8 \cdot 13$ and $3556 = 2^2 \cdot 7 \cdot 127$. (The prime $q = 12289$ does satisfy it, $12288 = 3 \cdot 2^{12}$, and $\mathrm{ord}_{257}(12289) = 256$, but it would require re-deriving the noise parameters, the security estimates and δ.) One must therefore use Karatsuba/Toom–Cook, Bluestein/Rader-type or lifted-modulus methods. No implementation or benchmark is presented and we make no performance claim.

9 Comparison with ML-KEM

	CLH-MAX (this work)	ML-KEM (FIPS 203)
Ring	$\mathbb{Z}_q[X]/\Phi_{257}$, $q \in \{2531, 3329, 3557\}$, field $\mathbb{F}_{q^{256}}$	$\mathbb{Z}_{3329}[X]/(X^{256}+1)$, 128 quadratic factors
Fast multiplication	no NTT for any of the three moduli	incomplete 7-layer NTT
Noise	ψ_2 throughout	ψ_3 in the key for ML-KEM-512
Generic MLWE estimate (key and message recovery)	Table 2: +0.2 bit ($k=2$, vs. weakest ML-KEM-512 instance), identical ($k=3$), -1.9 bits ($k=4$)	same estimator call
Worst-case motivation	LPR13, outside concrete regime	same
Standardisation / analysis	none (research proposal)	NIST standard, extensive analysis
δ	$\leq 2^{-133}$ for all three sets (heuristic, Table 3)	see FIPS 203

Table 5: Qualitative comparison; no security ranking is claimed.

10 Artifact availability

The source files of this version (LaTeX source, the Lean 4 development, the scripts of Appendix C and the raw estimator output) are archived on Zenodo:

- version 2.0.1 (this paper): `https://doi.org/10.5281/zenodo.23460069`
- all versions (always resolves to the latest): `https://doi.org/10.5281/zenodo.22883946`

Earlier Zenodo versions (v1 to 1.6) contain the previous version of this work, whose claims are withdrawn; version 2.0.0 used the modulus $q = 3329$ for all module ranks. See the file `ERRATA.md` in the repository.

The file `ClhMaxSecurity.lean` compiles against Mathlib v4.35.0 and formalises the game-hopping inequality underlying Theorem 7 for one ring component ($k = 1$): the left-or-right IND-CPA advantage is bounded by the sum of the advantages of four explicit distinguishers against (Ring-)LWE with one and two samples. It uses no hardness axiom and no `sorry`; `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. It does not formalise the hardness of (M)LWE, running times, or the module case $k > 1$.

11 Open problems

(i) A rigorous bound on δ (instead of the independence heuristic) and the effect of ciphertext compression on δ and on ciphertext size. (ii) Narrowing the residual gap of about two bits at $k = 4$, e.g. with other (η, q) or with different noise for secret and error. (iii) Efficient implementation: none of the moduli admits an NTT. (iv) Whether the field structure and the order-256 Galois action of R_q give any algebraic leverage against MLWE. (v) A machine-checked proof of Theorem 7 for $k > 1$ (the Lean file covers $k = 1$).

A Proof of Lemma 9

Coefficient i of $ab \bmod \Phi_{257}$ is $p_i - p_{256}$ with $p_i = \sum_{j+l\equiv i \ (257)} a_j b_l$, $0 \leq j, l \leq 255$. For $0 \leq i \leq 255$: pairs with $j + l = i$ number $i + 1$, and pairs with $j + l = i + 257$ number $255 - (i + 2) + 1 = 254 - i$

(for $i \le 254$) and 0 for $i = 255$. So p_i has 255 terms for $i \le 254$ and 256 terms for $i = 255$. For p_{256}: $j + l = 256$ gives 255 pairs ($j = 1, \ldots, 255$); $j + l = 513 > 510$ is impossible, so p_{256} has 255 terms. The index-pair sets of p_i and p_{256} are disjoint, hence the coefficient is a signed sum of $255 + 255 = 510$ products, or $256 + 255 = 511$ for $i = 255$. □

B Simulators for Theorem 7

```
B1(A, t*):                          # MLWE challenge: k samples, secret s
  (m0, m1) <- Adv.choose(pk=(A, t*)); b <- {0,1}
  r, e1 <- chi^k; e2 <- chi
  u = A^T r + e1; v = t*^T r + e2 + Enc1(m_b)
  b' <- Adv.guess(pk, (u, v)); return [b' == b]

B2(B, w):                           # MLWE challenge: k+1 samples, secret r
  parse B = [A^T ; t^T]             # A in R^{k x k}, t in R^k
  (m0, m1) <- Adv.choose(pk=(A, t)); b <- {0,1}
  u = w[1..k]; v = w[k+1] + Enc1(m_b)
  b' <- Adv.guess(pk, (u, v)); return [b' == b]
```

Real MLWE samples reproduce the corresponding game exactly; uniform samples reproduce the next game.

C Reproducibility scripts

Decryption failure probability (exact convolution under the independence heuristic; $e, s, r \leftarrow \psi_{\eta_1}$, $e_1, e_2 \leftarrow \psi_{\eta_2}$; threshold $\lfloor q/4 \rfloor$).

```
import numpy as np
from math import log2, comb
def cbd(eta):
    return np.array([comb(2*eta, i) for i in range(2*eta+1)], float)/4**eta, eta
def prod_pmf(x, y):
    p1, o1 = cbd(x); p2, o2 = cbd(y); d = {}
    for i, a in enumerate(range(-o1, o1+1)):
        for j, b in enumerate(range(-o2, o2+1)):
            d[a*b] = d.get(a*b, 0) + p1[i]*p2[j]
    off = max(abs(v) for v in d); arr = np.zeros(2*off+1)
    for v, p in d.items(): arr[v+off] += p
    return arr, off
def conv_pow(p, off, m):
    res, roff, base, boff = np.array([1.0]), 0, p.copy(), off
    while m:
        if m & 1: res = np.convolve(res, base); roff += boff
        m >>= 1
        if m: base = np.convolve(base, base); boff *= 2
    return res, roff
def delta(k, eta1, eta2, q):
    thr = q // 4
```

```
    pa, oa = prod_pmf(eta1, eta1); pb, ob = prod_pmf(eta1, eta2)
    d1, o1 = conv_pow(pa, oa, k*511); d2, o2 = conv_pow(pb, ob, k*511)
    d, o = np.convolve(d1, d2), o1 + o2
    p2, oo = cbd(eta2); d, o = np.convolve(d, p2), o + oo
    idx = np.arange(len(d)) - o
    t = d[np.abs(idx) >= thr].sum()
    return log2(t), log2(t) + 8           # per coefficient, union over 256
for k, e1, e2, q in [(2,2,2,2531), (3,2,2,3329), (4,2,2,3557), (4,2,2,3329)]:
    print(k, e1, e2, q, delta(k, e1, e2, q))
```

lattice-estimator runs (same call for all rows; key recovery has $m = n$, message recovery $m = 256(k+1)$).

```
from estimator import *
CBD = ND.CenteredBinomial
# name, k, q, eta_secret, eta_error
# (ML-KEM-512 is handled separately below: key psi_3/psi_3, message psi_3/psi_2)
sets = [("CLH-512", 2, 2531, 2, 2), ("CLH-768", 3, 3329, 2, 2),
        ("CLH-1024", 4, 3557, 2, 2), ("ML-KEM-768", 3, 3329, 2, 2),
        ("ML-KEM-1024", 4, 3329, 2, 2)]
for name, k, q, es, ee in sets:
    for kind, m in (("key", 256*k), ("message", 256*(k+1))):
        p = LWE.Parameters(n=256*k, q=q, Xs=CBD(es), Xe=CBD(ee), m=m,
                           tag=name + "-" + kind)
        print(name, kind); print(LWE.estimate(p))
for kind, es, ee, m in (("key", 3, 3, 512), ("message", 3, 2, 768)):
    p = LWE.Parameters(n=512, q=3329, Xs=CBD(es), Xe=CBD(ee), m=m,
                       tag="ML-KEM-512-" + kind)
    print("ML-KEM-512", kind); print(LWE.estimate(p))
```

References

[1] M. Albrecht, S. Bai, L. Ducas. A subfield lattice attack on overstretched NTRU assumptions. CRYPTO 2016.

[2] M. Albrecht, R. Player, S. Scott. On the concrete hardness of learning with errors. J. Math. Cryptology 9(3), 2015.

[3] J. Bos, L. Ducas, E. Kiltz, T. Lepoint, V. Lyubashevsky, J. Schanck, P. Schwabe, G. Seiler, D. Stehlé. CRYSTALS-Kyber: a CCA-secure module-lattice-based KEM. EuroS&P 2018.

[4] J. H. Cheon, J. Jeong, C. Lee. An algorithm for NTRU problems and cryptanalysis of the GGH multilinear map without a low-level encoding of zero. LMS J. Comput. Math. 19(A), 2016.

[5] R. Cramer, L. Ducas, C. Peikert, O. Regev. Recovering short generators of principal ideals in cyclotomic rings. EUROCRYPT 2016.

[6] R. Cramer, L. Ducas, B. Wesolowski. Short Stickelberger class relations and application to Ideal-SVP. EUROCRYPT 2017.

[7] D. Hofheinz, K. Hövelmanns, E. Kiltz. A modular analysis of the Fujisaki–Okamoto transformation. TCC 2017.

[8] H. Baan, S. Bhattacharya, S. Fluhrer, O. Garcia-Morchon, T. Laarhoven, R. Rietman, M.-J. O. Saarinen, L. Tolhuizen, Z. Zhang. Round5: Compact and fast post-quantum public-key encryption. PQCrypto 2019 (ePrint 2019/090).

[9] V. Lyubashevsky, C. Peikert, O. Regev. On ideal lattices and learning with errors over rings. J. ACM 60(6), 2013.

[10] NIST. FIPS 203: Module-Lattice-Based Key-Encapsulation Mechanism Standard. 2024.